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Modelling and Forecasting Seasonal Time Series using the ARMA-version of the Berlin Method

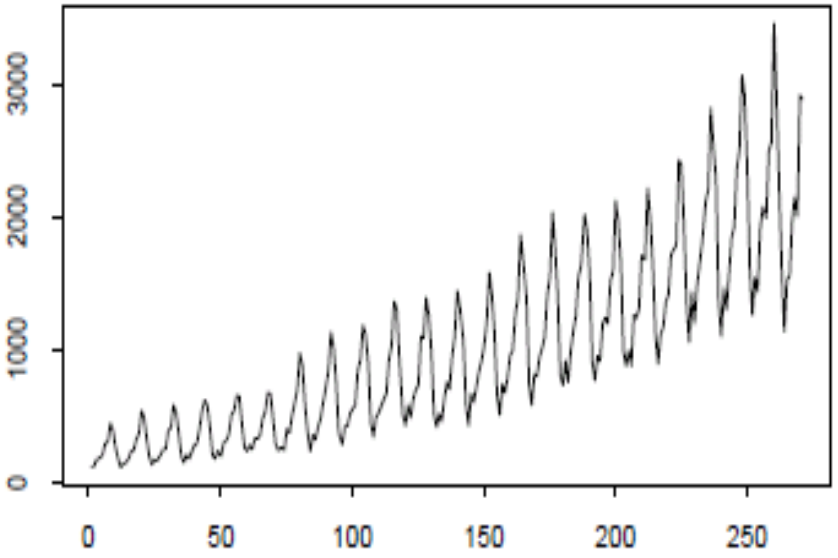
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Agenda

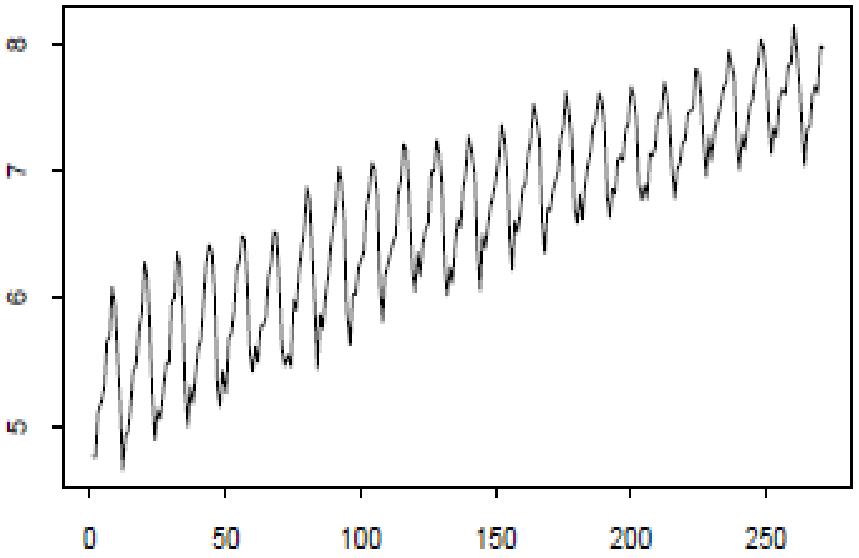
- Introduction
- Berlin Method
- Semiparametric Regression Model with ARMA Errors and Seasonality
- Bandwidth Selection: Plug-In
- Forecasting
- Application
- Conclusion

Introduction: Seasonal Time Series

- Many time series in the practice have a deterministic trend or a periodic component (called seasonality) due to economic development, business cycle or some natural facts, e.g. we do not eat so many ice creams in the Winter .
- An example:



Monthly UK-Tourism overseas expenditure, Jan 1980-Jul 2002



Log-transform of the original data

Berlin Method (Berliner Verfahren)

- In Germany, the decomposition and the seasonal adjustment of economic time series with the BV have a long tradition.
- In the late 1960s, BV was developed by the Berlin Technical University and the German Institute for Economic Research. Its fourth version is being applied by the German Federal Statistical Office since 1983.
- The method assumes a series to be additively composed by four components, namely trend, seasonal, calendar and residual.
  - trend fluctuations: technological progress, population growth
  - seasonal fluctuations: climatic conditions
  - calendar fluctuations: various weekdays, public holidays
  - residual fluctuations: strikes, natural disasters

Berlin Method

- The trend component is approximated by a polynomial of order 3, while the seasonal component is approximated by 11 trigonometric functions for monthly series(Eurostat.1998):

$$\hat{G}_t = \sum_{i=0}^{3} \hat{a}_i t^i \qquad \hat{S}_t = \sum_{i=1}^{6} \hat{b}_i \cos \lambda_i t + \sum_{i=1}^{5} \hat{c}_i \sin \lambda_i t$$

- **Advantages:**

  Speed of calculations is quick.

  Applying does not require user training or expert knowledge.

- **Disadvantages:**

  It is impossible for forecasting.

  The bandwidth is fixed and chosen in advance.

Semiparametric Regression Model with ARMA Errors and Seasonality

- An equidistant time series Y_t follows an additive components model:

$$Y_t = m(x_t) + Z_t = G(x_t) + S(x_t) + Z_t \quad x_t := (t - 0.5)/n$$

- The residual Z_t can be estimated by a stationary ARMA(p, q) model

$$Z_t = \beta_1 Z_{t-1} + \cdots + \beta_p Z_{t-p} + \alpha_1 \varepsilon_{t-1} + \cdots \alpha_q \varepsilon_{t-q} + \varepsilon_t$$

with summable autocovariances: $V = \sum_{k=-\infty}^{\infty} \gamma(k)$

- Around a point t_0 G can be locally modeled by a polynomial of order p assuming that it is at least (p+1)-times differentiable:

$$G(x_t) = \sum_{j=0}^{p} \beta_{1j}(t_0)(t - t_0)^j$$

- S can be locally modeled by a Fourier series:

$$S(x_t) = \sum_{j=1}^{q} [\beta_{2j}(t_0) \cos \lambda_j (t - t_0) + \beta_{3j}(t_0) \sin \lambda_j (t - t_0)]$$

Semiparametric Regression Model with ARMA Errors and Seasonality

- The locally weighted estimators are obtained by solving the least square problem:

$$\sum_{t=1}^{n}\Big\{Y_t-\sum_{j=0}^{p}\beta_{1j}(t_0)(t-t_0)^j - \sum_{j=1}^{q}[\beta_{2j}(t_0)\cos\lambda_j(t-t_0)+\beta_{3j}(t_0)\sin\lambda_j(t-t_0)]\Big\}K\left(\frac{t-t_0}{h}\right)\Rightarrow \min$$

leads to the solutions:

$$\hat{\beta}(t_0)=(X^TKX)^{-1}X^TKy$$

$$\hat{m}(x_t)=(e_1^T,\Phi_s^T)(X^TKX)^{-1}X^TKy=:w^Ty$$

$$\hat{G}(x_t)=(e_1^T,0^T)(X^TKX)^{-1}X^TKy=:w_1^Ty$$

$$\hat{S}(x_t)=(0^T,\Phi_s^T)(X^TKX)^{-1}X^TKy=:w_2^Ty$$

Asymptotic Properties

- A 1: h $\longrightarrow$ 0 and nh$\longrightarrow$ ∞ as n$\longrightarrow$ ∞.

  A 2: G is at least k(=p+1) times continuously differentiable.

  A 3: S is exactly periodic with period s.

- MASE: $$MASE(\hat{m}) = \frac{V}{nh}\{R(K_P) + (s-1)R(K)\}$$
$$+ \frac{1}{(k!)^2}\left\{\int \{G^{(k)}(x)\}^2 dx \left[\int u^k K(u)du\right]^2\right\} h^{2k}$$

  The asymptotically optimal bandwidth, which minimizes MASE is given by

$$h_{opt} = \left(\frac{(k!)^2}{2k} \frac{V\{R(K_p) + (s-1)R(K)\}}{\int \{G^{(k)}(x)\}^2 dx \left\{\int u^k K_p(u)du\right\}^2} \right)^{1/(2k+1)} n^{-1/(2k+1)}$$

  with unknown parameters : V and $I = \int \{G^{(k)}(x)\}^2 dx$

Bandwidth Selection: Plug-In

- Estimation of variance:
  - i.i.d. errors(4 iterations): seasonal-difference-based method
  - ARMA errors(> 4 iterations):

$$\hat{V} = \sum_{k=-k_{\max}}^{k_{\max}} \hat{\gamma}(k) \quad \text{where} \quad k_{\max} = [n^{\frac{1}{5}}] + 1$$

$$\hat{\gamma}(k) = \frac{1}{n}\sum_{t=1}^{n-k}(x_t - \bar{x})(x_{t+k} - \bar{x}) \qquad k = 0,1,\ldots,k_{\max}$$

- Estimation of I:

  k-th derivative of G can be estimated with a polynomial order of p_I and a bandwidth h_I with $p_I > k$ and $p_I - k$ odd. Usually, $p_I = k+1$.

$$\hat{G}^{(k)}(x_t) = k!(e_{k+1}^T, 0^T)(X^T K X)^{-1} X^T K y =: (w^k)^T y$$

  Then I can be estimated by

$$\hat{I}[G^{(k)}(x; h_I)] = \frac{1}{n}\sum_{i=1}^{n}\left\{\hat{G}^{(k)}(x_i; h_I)\right\}^2$$

Forecasting

- The point forecasts model is defined by(k=1,…,12 months)

$$\hat{Y}_{n+k} = \hat{G}_{n+k} + \hat{S}_{n+k} + \hat{Z}_{n+k}$$

- The forecasts of trend: increase according to linear extrapolation

$$\hat{G}_{n+k} = \hat{G}_n + k\Delta G_n \qquad \Delta G_n = \hat{G}_n - \hat{G}_{n-1}$$

- The forecasts of seasonality: use estimates of the last 12 months

$$\hat{S}_{n+k} = \hat{S}(n-s+k)$$

- The forecasts of residual:

$$k=1, \hat{Z}_{n+1} - \hat{\mu} = \beta_1(Z_n - \hat{\mu}) + \cdots + \beta_p(Z_{n+1-p} - \hat{\mu})$$

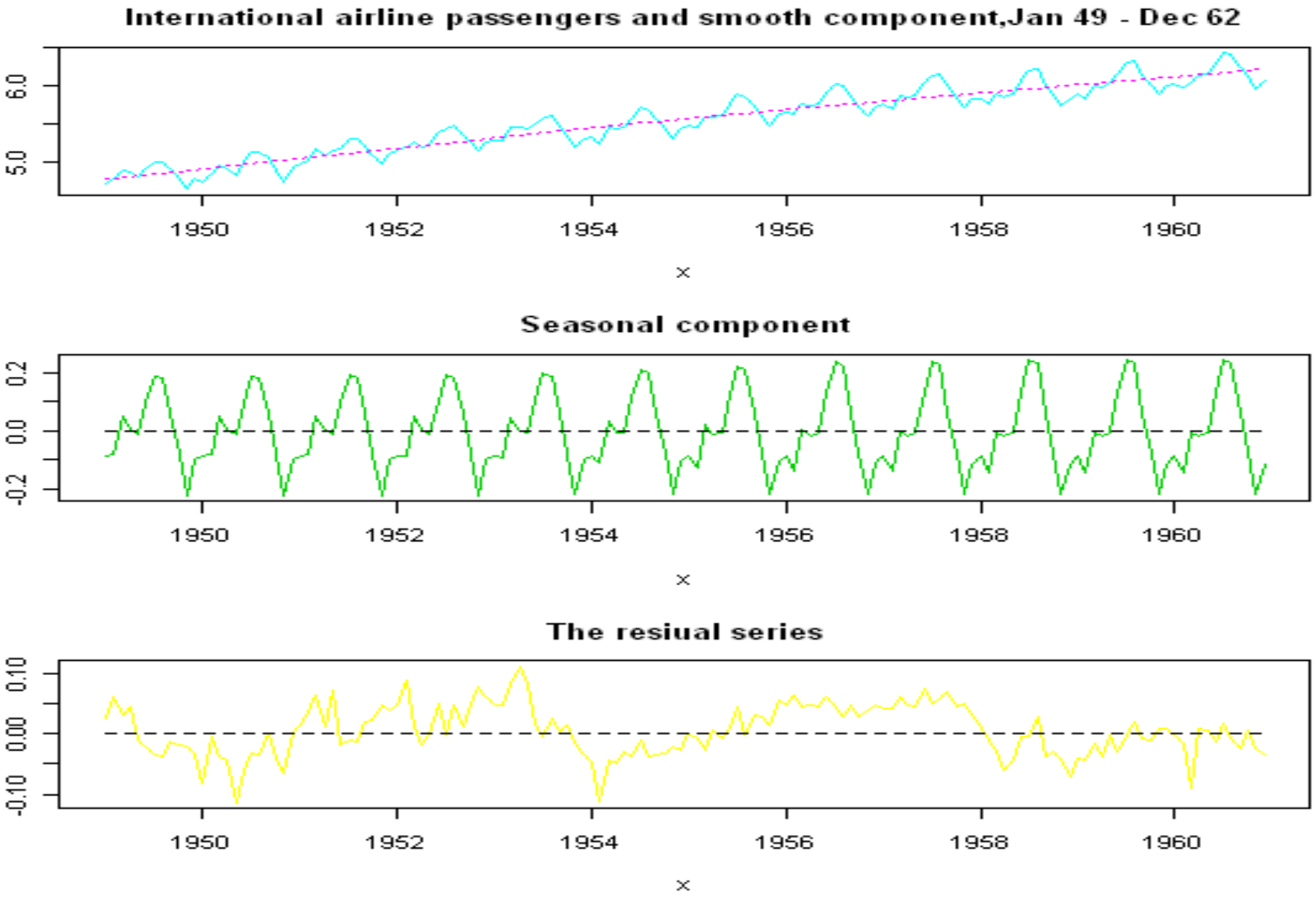
$$k=2, \hat{Z}_{n+2} - \hat{\mu} = \beta_1(\hat{Z}_{n+1} - \hat{\mu}) + \beta_2(Z_n - \hat{\mu}) \cdots + \beta_p(Z_{n+2-p} - \hat{\mu})$$

$$k > p, \hat{Z}_{n+k} - \hat{\mu} = \beta_1(\hat{Z}_{n+k-1} - \hat{\mu}) + \cdots + \beta_p(Z_{n+k-p} - \hat{\mu})$$

Application

- *Monthly international airline passengers from January 1949 to December 1962 (Point forecasts for one year)*
- *Australia employed persons from February 1978 to April 1991 (Point forecasts for one year)*
- Total takings from accommodation from January 1980 to June 1995
- Monthly Australian imports from Japan from July 1965 to October 1993
- Total number of room nights occupied from January 1980 to June 1995
- Monthly Australian sales of redwine from January 1980 to July 1995
- US monthly sales of chemicals from January 1971 to December 1991

Example 1


International airline passengers and smooth component,Jan 49 - Dec 62
6.0
5.0
1950
1952
1954
1956
1958
1960
x
Seasonal component
0.2
0.0
-0.2
1950
1952
1954
1956
1958
1960
x
The resiual series
0.10
0.00
-0.10
1950
1952
1954
1956
1958
1960
x

Number of observations	144
Number of iteration	10
h	56
2h+1	113
VarD	0.0004360177
VarS	0.008004573

p \ q	0	1
0	-508.9748	-559.2373
1	-604.3391	-608.4435
2	***-608.7617***	-603.8375
3	-603.8309	-603.6565
4	-598.8709	-598.8729
5	-596.8050	-591.8418

	ar1	ar2	intercept
Coefficients	0.5294	0.2523	0.0047
s.e.	0.0803	0.0809	0.0104

BIC: p=2, q=0.
AR(2) model

- The selected best model is AR(2) model:

$$\hat{Z}_t - 0.0047 = 0.5294(\hat{Z}_{t-1} - 0.0047) + 0.2523(\hat{Z}_{t-2} - 0.0047) + \varepsilon_t$$

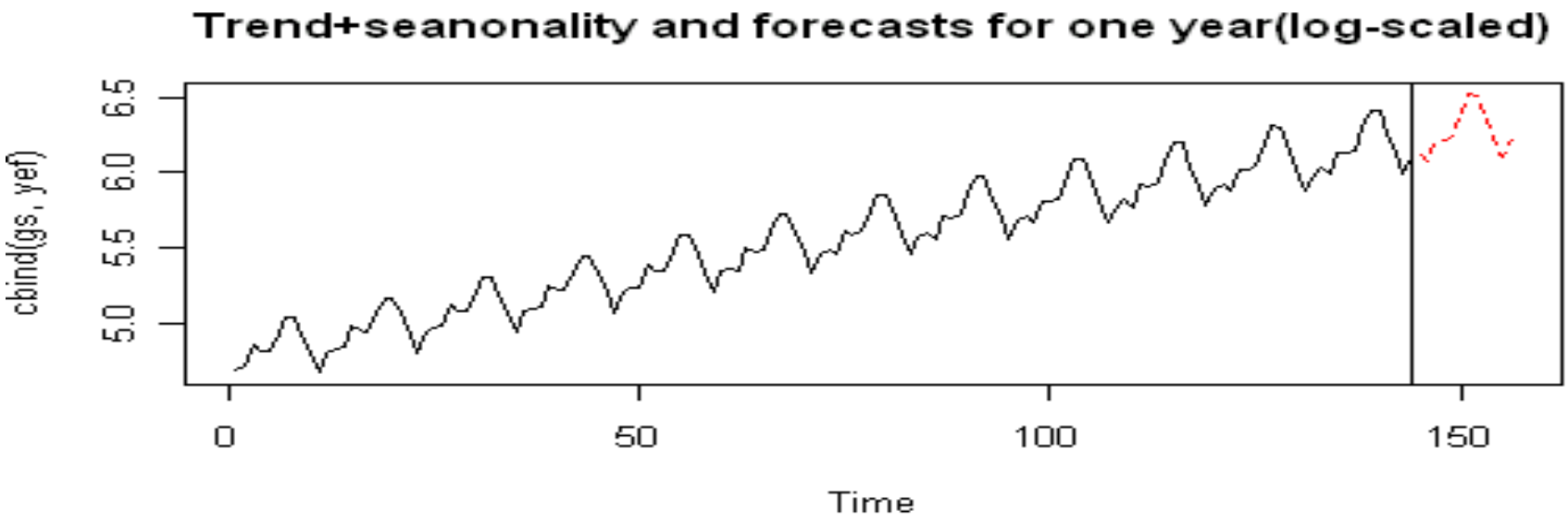
- The forecasts model: $\hat{Y}_{n+k} = \hat{G}_{n+k} + \hat{S}_{n+k} + \hat{Z}_{n+k}$
- For k=1: $\hat{G}_{n+1} = \hat{G}_n + (\hat{G}_n - \hat{G}_{n-1}) = 6.2179 + (6.2179 - 6.2090) = 6.2268$

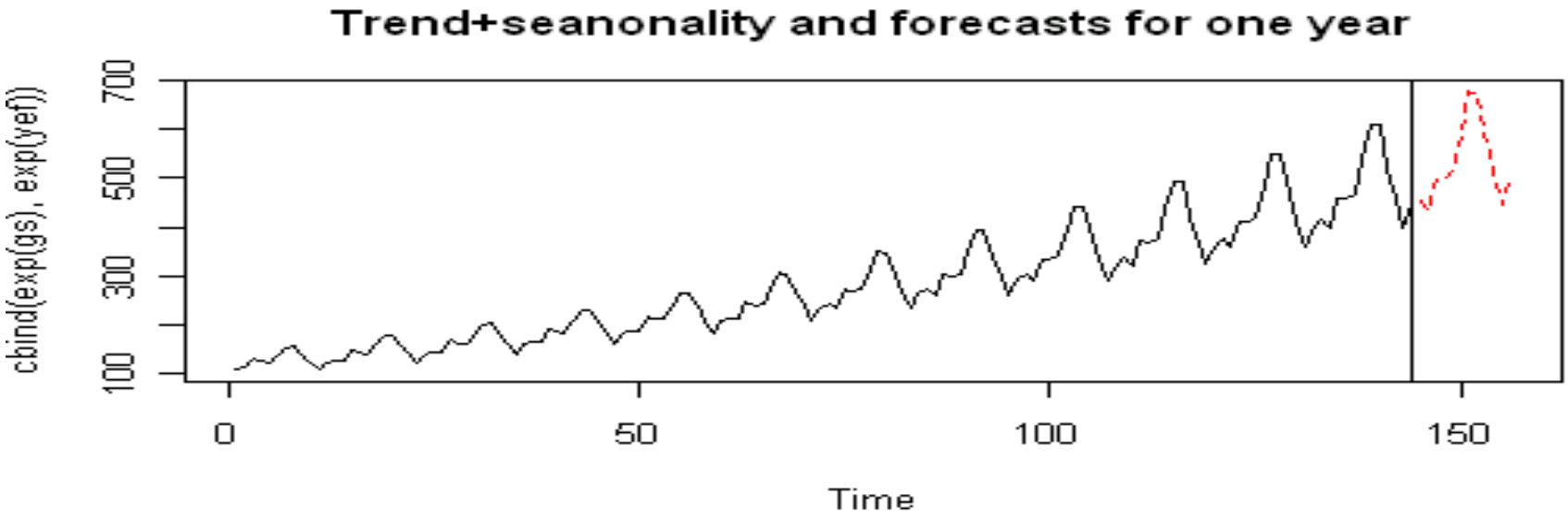
$$\hat{S}_{n+1} = \hat{S}(n-12+1) = \hat{S}(133) = -0.0857$$

$$\begin{aligned}\hat{Z}_{n+1} &= 0.5294(Z_n - 0.0047) + 0.2523(Z_{n+1-2} - 0.0047) + 0.0047 \\ &= 0.5294(-0.0346 - 0.0047) + 0.2523(-0.0252 - 0.0047) + 0.0047 \\ &= -0.0236\end{aligned}$$

	K=1	**K=2**	**K=3**	**…**	**K=12**
$\hat{G}_{n+k}$	6.2268	6.2357	6.2446	…	6.3247
$\hat{S}_{n+k}$	-0.0857	-0.1428	-0.0089	…	-0.1149
$\hat{Z}_{n+k}$	-0.0236	-0.0202	-0.0157	…	0.0007
Σ	6.1174	6.0727	6.2201	…	6.2106

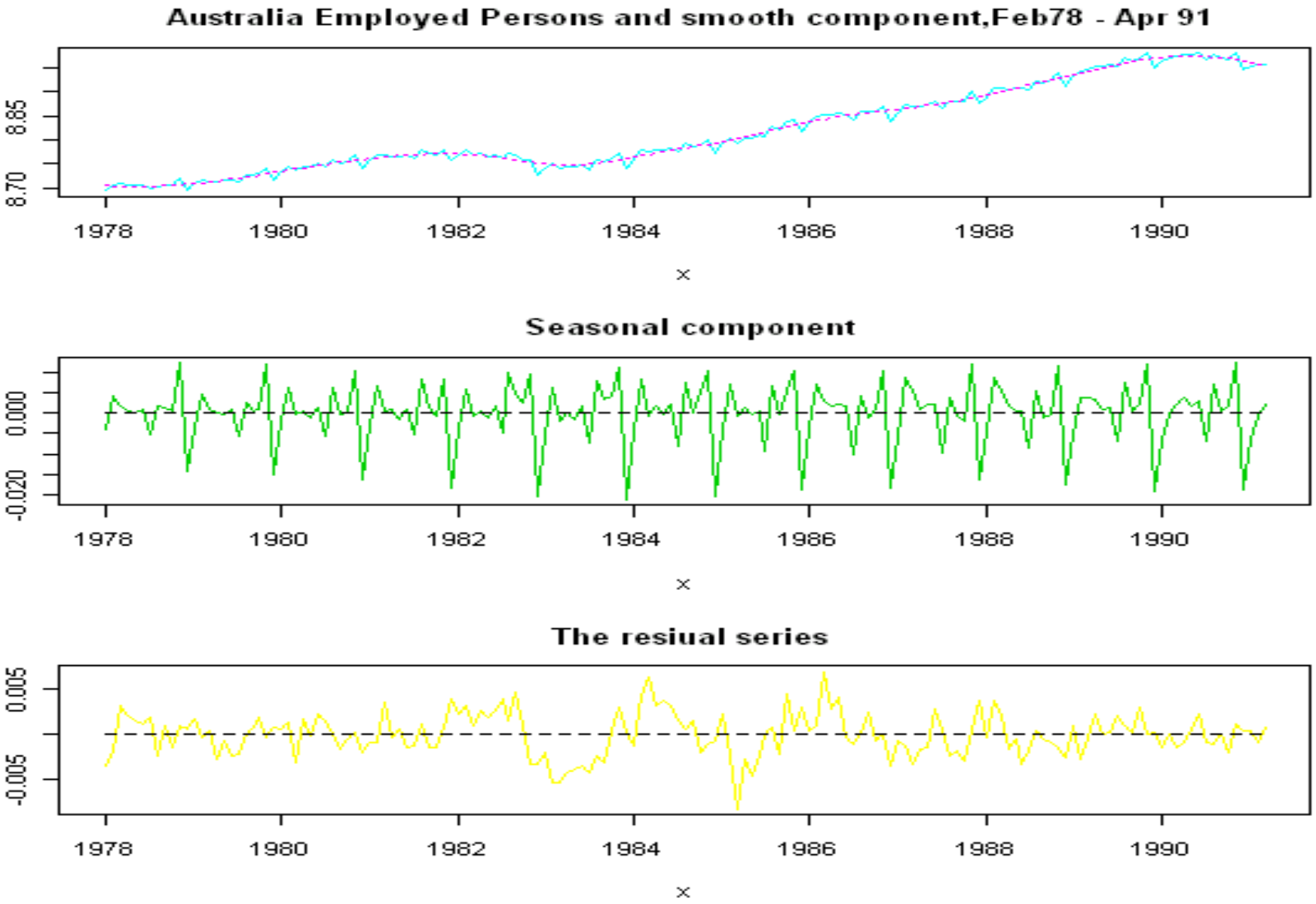
Forecasts of trend, seasonality, residual and the sum of them for one year





The sum of trend and seasonality with forecasts for one year (log-scaled and exponential)

Example 2


Australia Employed Persons and smooth component,Feb78 - Apr 91
8.70
8.85
1978
1980
1982
1984
1986
1988
1990
x
Seasonal component
-0.020
0.000
1978
1980
1982
1984
1986
1988
1990
x
The resiual series
-0.005
0.005
1978
1980
1982
1984
1986
1988
1990
x

Number of observations	159
Number of iteration	8
h	18
2h+1	37
VarD	6.961188e-06
VarS	1.722117e-05

p \ q	0	1
0	-1469.673	-1489.311
1	***-1501.066***	-1500.433
2	-1500.912	-1495.886
3	-1495.902	-1490.860
4	-1490.960	***-1505.272***
5	-1486.099	-1502.386

	ar1	**intercept**
Coefficients	0.4544	0e+00
s.e.	0.0708	3e-04

BIC: p=4, q=1: ARMA(4,1) model

p=1, q=0: AR(1) model

- The fitted AR(1) model is defined by:

$$\hat{Z}_t = 0.4544\hat{Z}_{t-1} + \varepsilon_t$$

➢The forecasts model: $\hat{Y}_{n+k} = \hat{G}_{n+k} + \hat{S}_{n+k} + \hat{Z}_{n+k}$

➢For k=1:

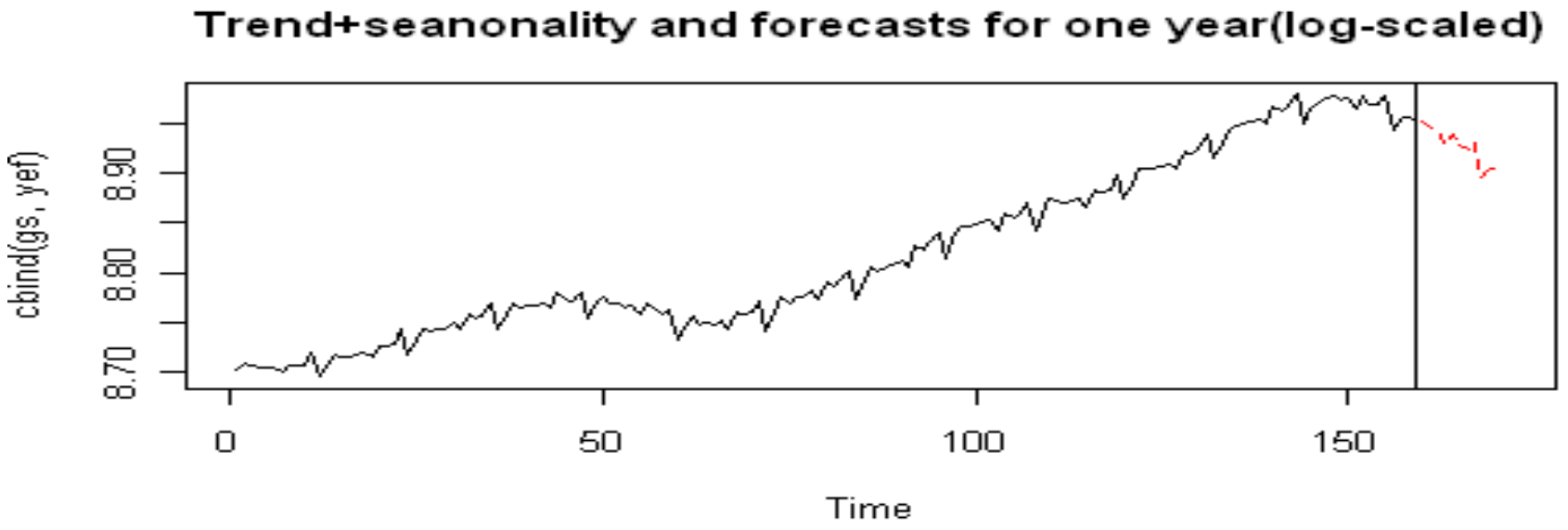
$$\hat{G}_{n+1} = \hat{G}_n + (\hat{G}_n - \hat{G}_{n-1}) = 8.9524 + (8.9524 - 8.9566) = 8.9482$$

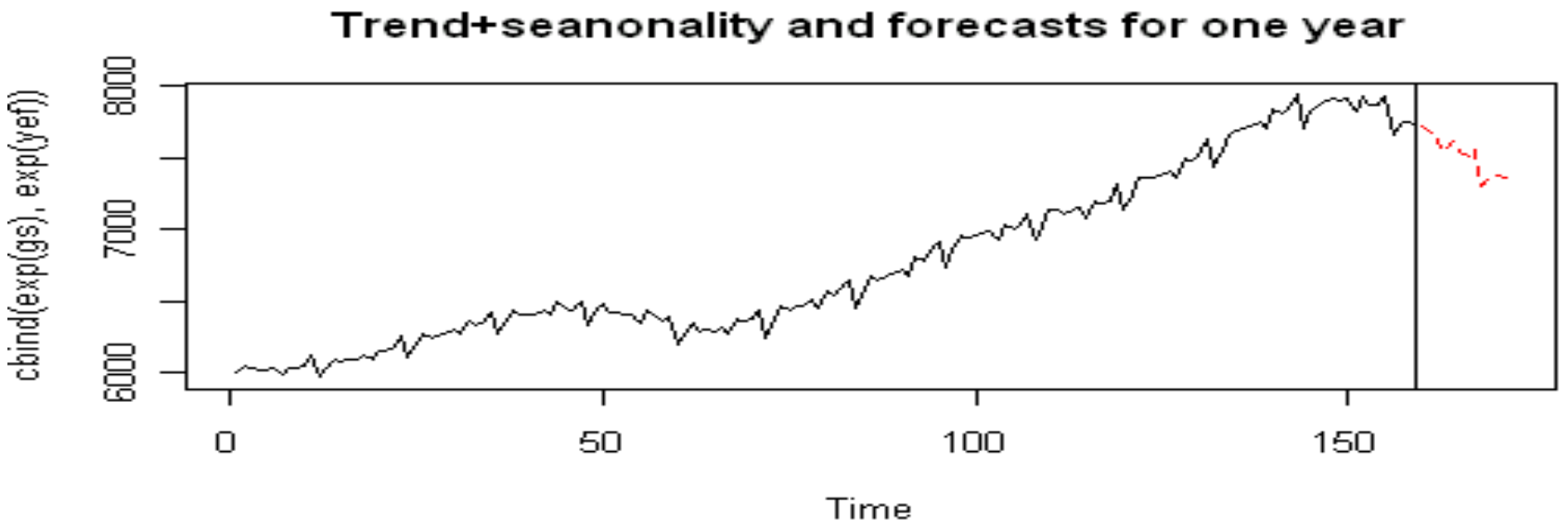
$$\hat{S}_{n+1} = \hat{S}(n-12+1) = \hat{S}(148) = 0.0040$$

$$\hat{Z}_{n+1} = 0.4544 Z_n = 0.4544 \times 0.0009 = 0.0004$$

	K=1	**K=2**	**K=3**	**…**	**K=12**
$\hat{G}_{n+k}$	8.9482	8.9440	8.9398	…	8.9020
$\hat{S}_{n+k}$	0.0040	0.0016	0.0031	…	0.0023
$\hat{Z}_{n+k}$	0.0004	0.0002	6.1060e-05	…	-2.1354e-05
Σ	8.9525	8.9457	8.9429	…	8.9043

Forecasts of trend, seasonality, residual and the sum of them for one year





The sum of trend and seasonality with forecasts for one year (log-scaled and exponential)

Conclusion

- Decomposing seasonal time series into trend, seasonality and residual has a long tradition.
- The trend and seasonality can be estimated by locally weighted regression, while the residual can be modeled by an ARMA process, so the whole model is said to be a semiparametric model.
- The asymptotically optimal bandwidth is obtained by minimizing the mean averaged squared error.
- Another focus is the short-term forecasting. Trend increases according to linear extrapolation, seasonality follows the last 12 months estimates, and residuals are based on the point predictions procedure for ARMA model.